Changes in Mother–Infant Relationship after Separation in *Rhesus* Monkeys

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Infant distress after mother-infant separation depends greatly on the re-establishment of maternal care on reunion.

IF separated from their mothers for a few days, infant *Rhesus*[1] monkeys and other macaques[2] show increased distress calling, and reduced locomotor activity and other symptoms of behavioural depression. The symptoms may persist for weeks after reunion with the mother, and infants which have had such a separation experience differ from non-separated controls in their responses to strange objects months and even years later[3]. The trauma of the separation experience itself may not be the only or even the major factor in producing these effects. The disturbance to the mother-infant relationship which results from the separation seems to be of considerable importance.

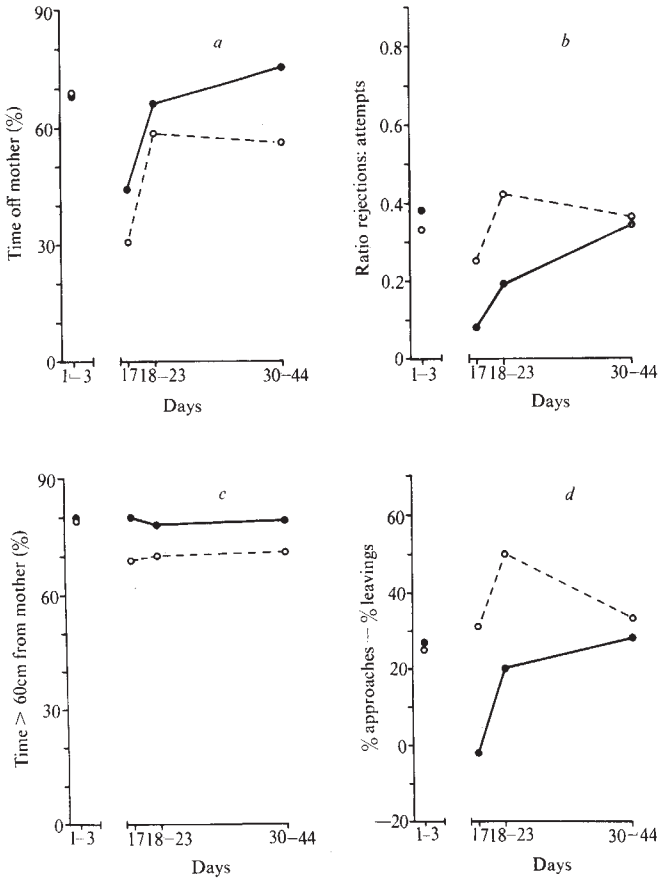


Fig. 1 Effects of 13 day separation on mother-infant relations. Separation was made on day 3. *a*, Number of half minutes in which infant was off mother as percentage of number observed. *b*, Ratio of number of times infants' attempts to gain nipple were rejected by mother to total number of attempts. *c*, Number of half minutes in which infant was >60 cm from mother as percentage of number in which off her. *d*, Difference between percentage of approaches due to infant and percentage of leavings due to infant. ○ - - - ○, Mother removed; ●—●, infant removed.

Mother-Infant Relationship after Reunion

In experiments involving removals of mothers from sixteen 21 to 32-week-old infants for six days[4] and from five 30 to 42-week-old infants for thirteen days[5], the mothers acceded only partially to the increased demands for nipple contact made after reunion. Over this age range non-separated infants spend progressively less time on their mothers. Separated infants spent more time on their mothers after reunion than before ($P<0.01$, Wilcoxon 2-tailed test, for the six day group and 5/6, *n.s.*, for the thirteen day group, for the day of reunion; $P<0.01$ and 5/6, *n.s.*, for the next week: in one of the 13 days group relations with the mother were not re-established for over a week), and this occurred in the absence of any significant reduction in the proportion of the infant's attempts to gain the nipple that were rejected by the mother[6]. Indeed, the proportion of nipple contacts terminated by the mother was markedly greater after reunion than before ($P<0.02$ for both day of reunion and the next week for the 13 day group: adequate data were not available for most 6 day animals, but similar trends were apparent). In 13 day animals median increased from 0.48 pre-separation to 0.94 in second week after reunion, compared with a mean change of +0.01 ($N=14$) in non-separated controls). By rejection of a high proportion of the infants' attempts to gain the nipple (Fig. 1) and termination of a high proportion of nipple contacts the mothers play a large part in determining the nature of the mother-infant interaction.

A similar conclusion is indicated by the proportion of time off their mothers that the infants spent out of arm's reach (>60 cm). After reunion this was reduced in nearly all infants $P<0.01$ and 5/6, *n.s.*, in the two experiments for the day of reunion; $P<0.01$ and 5/6, *n.s.*, for the next week). The infant's role in maintaining proximity to its mother can be assessed by the difference between the percentage of approaches due to movement by the infant and the percentage of leavings due to movement by the infant ($\%Ap-\%L$), where approaches and leavings occur when the infant's position changes with respect to a circle of 60 cm radius round the mother[7]. Positive values indicate that the infant is responsible for a higher proportion of approaches than it is of leavings, and thus is primarily responsible for maintaining proximity. After reunion this index tended to increase (8/10, *n.s.*, and 3/6, *n.s.*, on day of reunion; 12/16, $P<0.02$ and 5/6, *n.s.*, in next week). Thus the decrease in time that the infants spent at a distance from their mothers occurred even though the mothers were playing a smaller role in maintaining mutual proximity.

We conclude that after temporary removal of the mothers, the mothers play a large part in limiting the closeness of the relationship after reunion.

Distress shown by the Infant after Reunion

The degree of distress shown by infants during separation and after reunion varies markedly between individuals. For the infants whose mothers were removed for 6 days, distress was measured by frequency of distress calling, the frequency with which they sat doing nothing, and the inverse of their locomotor activity. This distress index was found to be significantly correlated with rejection of attempts to gain the nipple and maintenance of proximity to the mother. Rank order correlation coefficients between pre-separation frequency of rejections and distress index for day of reunion, next week and days 13–27 after reunion respectively +0.18, *n.s.*; +0.59, $P<0.05$; and +0.43, $P<0.05$. Correlations between measure of

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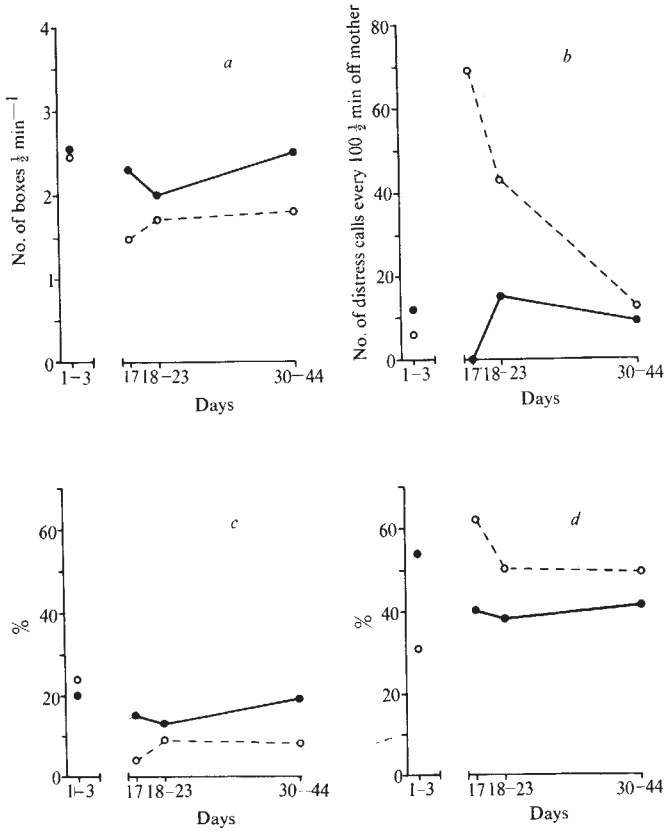


Fig. 2 Effects of 13 day separation from mother on locomotor behaviour of infant *Rhesus* monkeys when off mothers. ●—●, Median for infant-removed infants; ○ - - - ○, that for mother-removed infants. *a*, Separation was made on day 3, number of sections of cage entered per half minute off mother. *b*, Frequency of distress calls. *c*, Number of half minutes in which infant entered more than 5 sections of cage, as a percentage of number in which it entered more than one. *d*, Number of half minutes in which infant sat doing nothing as a percentage of number off mother.

infant's role in maintaining proximity and distress index respectively +0.30, *n.s.*; +0.64, $P<0.01$; and 0.41, *n.s.* The distress index immediately after reunion was more closely related to the pre-separation mother-infant relationship than post-separation, but by 2–4 weeks after separation the situation was reversed[8]. Although these correlations do not prove that individual differences in the mother-infant relationship determined differences in infant distress, this seems highly probable.

Infant-removed and Mother-removed Infants

In the previous experiments involving temporary removal of the mother, the infant remained in the same physical surroundings and with the same social companions. If the infant were removed from the mother, the infant should be more severely affected, for the trauma of the confinement in a strange situation would be added to that of separation. Data on 5 infants removed from their mothers for 13 days (infant-removed infants) were compared with those on 6 whose mothers were removed (mother-removed infants). In each case data were collected for 3 days before separation, mother and infant were separated on day 4 and reunited on day 17, and subsequent data were pooled for days 18, 19, 21 and 23 and for days 30, 37 and 44. Differences between the physical conditions of the cages in which these infant-removed infants were kept and the home pens in which the mother-removed infants remained prevented precise comparisons, but it did seem that the infant-removed infants were the more disturbed. After reunion, however, the reverse was the case[9]. The infant-removed infants showed more locomotor activity ($P=0.04$, 0.03 and 0.01 for days 17, 18–23 and 30–44, respectively, Mann Whitney 2-tailed), were more active when active (*n.s.*, *n.s.* and $P=0.01$), and gave fewer distress calls ($P=0.004$, 0.03 and *n.s.*), than infants whose mothers had been removed (Fig. 2). They also engaged in significantly more social play.

The small degree of distress shown by the infant-removed infants was accompanied by a smaller disruption of mother-infant relations (Fig. 1). The time spent off the mother, greater in the infant-removed infants than in the mother-removed ones (*n.s.*, *n.s.* and $P=0.05$ on days 17, 18–23 and 30–44 respectively), was associated with a lesser relative frequency of rejections ($P<0.06$, 0.06, *n.s.*, Sign test). There was, however, no significant change in the proportion of nipple contacts made or broken by the infants. The improved relationship was thus primarily due to greater receptiveness by the mothers to the infants' demands for nipple contacts. By contrast, in the mother-removed group the relative frequency of rejections and the infant's role in making nipple contacts changed little, but there was a marked decrease in the proportion of nipple contacts broken on the infant's initiative ($P<0.02$, 0.02 and 0.02). The mothers not becoming more receptive played a greater role in terminating nipple contacts.

Similar considerations apply to the relative roles of mother and infant in maintaining proximity. The infant-removed infants spent a slightly higher proportion of their time off their mothers at a distance from her than did the mother-removed infants (*n.s.*, *n.s.* and $P=0.01$). The infants' role in maintaining proximity tended to be greater in the mother-removed infants (*n.s.*) than before separation, and less ($P=0.06$, *n.s.*, *n.s.*) in the infant-removed infants, though the inter-group differences did not reach significance.

Thus the smaller distress shown by the infant-removed infants after reunion was associated with a smoother re-establishment of mother-infant relations, and the latter was due to a readier acceptance by the mothers of the demands for nipple contact and proximity made by the infants after reunion. Presumably the mothers, whose role in determining the nature of the mother-infant relationship after separation has already been emphasized, suffer less stress if their infants are merely removed than if they are both separated from their infants and placed in a strange environment. These findings support the view that the disturbance to the mother-infant relationship caused by the separation episode is an important factor in the distress shown by the infant in the post-separation period.

Two further points must be made. First, this suggestion does not imply that separation *per se* has no subsequent effects. The infants show marked distress during separation, and it would be surprising if this had no later consequences. Second, our opinion is that there are no human parallels, though no data exist. Disturbance to the relationship is found in both species, however. While the view that separation from the mother may have damaging effects on the development of the human child[10] is well substantiated[11], there is growing evidence that the crucial factor in separation is the deprivation of maternal care which may, but need not, accompany it[12].

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